Disentangling Conventional and Unconventional Monetary Policy Shocks in an Emerging Market

February 2026

Carlos Alba[1]

Abstract

This paper analyzes the macroeconomic effects of innovations to the central bank's target policy rate and forward guidance (FG) in an emerging market economy. For each policy tool, we disentangle pure monetary policy innovations from central bank information (CBI) and central bank response-to-news (RN) components embedded in high-frequency monetary policy surprises. We show that, when CBI and RN components are not isolated, contractionary target-rate shocks exhibit the familiar output, price, and exchange-rate puzzles and are followed by a counterintuitive rise in inflation expectations, while contractionary FG shocks appear to generate disproportionately large and fast real effects. Once CBI and RN components are isolated, the estimated effects of contractionary target-rate shocks become broadly consistent with standard monetary transmission mechanisms, featuring delayed contractions in output and prices, an appreciation of the domestic currency, and well-anchored inflation expectations. In turn, the effects of FG shocks become more conservative and broadly similar to those of pure target-rate shocks. Notably, RN components play asymmetric contamination roles across policy instruments: they tend to attenuate the contractionary effects of target-rate shocks, but amplify and accelerate the estimated effects of FG innovations.

Keywords Monetary Policy; High-Frequency Identification; Central Bank Information; Response to News; Forward Guidance; Bayesian VAR; Emerging Market Economies

JEL codes E43; E52; E58; F42

GitHub Repo: https://github.com/Calbafa/calbafa606
PDF: HAFiscal.pdf

Draft (REMARK) — Repository: `calbafa606`

We thank Marco Bassetto, Laurence Ball, Francesco Bianchi, Christopher Carroll, Stelios Fourakis, Raul Ibarra, David Rappoport, Jonathan Wright, as well as participants at the Macro/Finance Seminar at Johns Hopkins University, for helpful comments and suggestions. The views expressed in this paper are those of the author and do not necessarily reflect those of Johns Hopkins University.

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1 Introduction

High-frequency changes in interest rates around central bank announcements—commonly referred to as monetary policy or interest rate surprises—have become a widely used proxy for monetary policy (MP) shocks. A growing literature combines these surprises with vector autoregressive (VAR) models or local projections to analyze the macroeconomic effects of monetary policy, primarily in advanced economies (AEs). Applying this methodology to emerging market economies (EMEs), however, poses additional challenges, partly reflecting the more limited availability and shorter span of high-frequency financial data.

Although recent contributions have extended high-frequency identification (HFI) techniques to some EMEs, a number of studies document puzzling responses to contractionary MP shocks, including exchange-rate depreciation, rising prices, and increases in inflation expectations. These findings are difficult to reconcile with standard macroeconomic models and raise important questions about the transmission and identification of MP shocks in EMEs.

A potential explanation for these puzzling findings is that interest rate surprises may not exclusively capture exogenous deviations from a systematic policy rule. If market participants are imperfectly informed about the state of the economy, these surprises may be contaminated by *central bank information* (CBI) effects (Romer and Romer, 2000; Jarociński and Karadi, 2020). Central banks may convey new information about the economic outlook in their announcements while simultaneously adjusting the policy stance in response. The resulting interest rate surprises then partly reflect an endogenous policy response to economic conditions, potentially biasing the transimission of MP shocks.

An additional source of contamination arises from *market misperceptions about the central bank's reaction function*. In this case, interest rate surprises may reflect an unexpectedly aggressive response of policymakers to observable developments, or *central bank response-to-news* (RN) effects, as emphasized by Bauer and Swanson (2023a, 2023b). As shown by these authors, failing to control for these effects may also bias the transmission of MP shocks.

Recent work for the U.S. by Jarociński and Karadi (2025) jointly analyzes the role of both contamination channels. Their results indicate that CBI effects substantially attenuate the transmission of monetary policy, while RN effects appear to play a more limited role. We extend this analysis to an EME and jointly examine how CBI and RN components shape the transmission of two types of interest rate surprises. Specifically, we distinguish between surprises to the target policy rate, representing conventional MP shocks, and surprises to the expected near-term policy path, capturing unconventional or forward-guidance (FG) shocks.

Our analysis shows that both CBI and RN components appear to play an important role in attenuating the effects of target-rate surprises and in accounting for several of the puzzles documented in the EME literature. For FG surprises, however, RN components appear to play a more prominent role. Notably, RN contamination seems to operate asymmetrically across policy instruments: it weakens the transmission of target-rate

shocks but disproportionately amplifies and accelerates the estimated effects of FG innovations.

Focusing on Mexico as an ideal case study, we begin by computing high-frequency changes in interest rate swaps at different maturities around monetary policy announcements.[1] Following the standard approach of Gürkaynak et al. (2005), we summarize these variations into a target-rate surprise, capturing unexpected changes in the policy rate, and an FG surprise, reflecting revisions to the expected near-term policy path.

Following the logic of Bauer and Swanson (2023a, 2023b), we then regress each policy surprise on a set of macroeconomic and financial variables observable to market participants prior to the announcement. The fitted value captures the portion of the surprise that may reflect an unexpectedly aggressive response of the central bank to observable economic developments. The residual isolates the unpredictable part of the surprise and therefore provides a cleaner proxy for exogenous monetary policy innovations.

In a second step, we exploit the high-frequency response of stock prices around policy announcements to further refine this decomposition and distinguish between the three shocks of interest for each surprise. We then trace their dynamic effects using a monthly Bayesian VAR estimated over the period January 2002 to December 2024.

In particular, a *pure MP shock* is characterized by a negative co-movement between the residual component of the interest rate surprise and the stock price surprise. This pattern is consistent with an unexpected policy tightening that raises discount rates and lowers expected future cash flows, leading to an immediate decline in equity prices.

An *RN shock*, in turn, is characterized by a negative co-movement between the fitted component of the interest rate surprise and the stock price surprise. This configuration captures episodes in which the central bank responds more aggressively than markets anticipated to publicly available information. Because that information was largely incorporated into expectations prior to the announcement, the only new element revealed within the announcement window is a stronger-than-expected policy response, which depresses equity prices through standard discount-rate and cash-flow channels.

Finally, a *CBI shock* is associated with a positive co-movement between the interest rate surprise and the stock price surprise. This pattern may arise when the central bank reveals favorable new information about economic fundamentals while simultaneously tightening policy. In such cases, equity prices increase in response to improved growth prospects and higher expected future cash flows, despite the policy tightening.

Our main results indicate that CBI and RN components tend to attenuate—and even reverse—the effects of target-rate surprises. When these components are not disentangled, impulse responses to contractionary target-rate shocks exhibit the well-known output, price, and exchange-rate puzzles, with inflation expectations rising on impact. Once the CBI and RN components are isolated, however, these puzzles largely disappear: output and prices display a U-shaped response with a delayed trough, the

[1] Mexico provides an ideal case study given its inflation-targeting regime, a flexible and market-determined exchange rate, and relatively deep and liquid financial markets, which allow the measurement of monetary policy surprises across different maturities—an essential feature for distinguishing between conventional and unconventional MP shocks.

domestic currency appreciates against the U.S. dollar, and inflation expectations remain well anchored. In addition, stock prices decline, and both the one-year and the ten-year government bond yields increase, with the long-term yield rising by a smaller amount.

In line with these results, favorable CBI shocks identified from target-rate surprises are followed by increases in interest rates, an expansion in output, higher prices, and rising inflation expectations. Notably, the Mexican peso depreciates against the U.S. dollar in response to these shocks, a pattern that may be consistent with upward revisions in inflation expectations, as emphasized by Gürkaynak et al. (2021) and Stavrakeva and Tang (2019). These responses are consistent with an interpretation in which the central bank reveals favorable information about the economic outlook while simultaneously tightening policy to lean against emerging inflationary pressures, potentially operating through aggregate demand effects and exchange-rate pass-through.

In turn, hawkish RN shocks identified from target-rate surprises are followed by increases in interest rates, an expansion in output, higher prices, rising inflation expectations, and a depreciation of the domestic currency. These dynamics closely resemble those observed following favorable CBI shocks and are in line with Bauer and Swanson (2023a, 2023b). In general, they are consistent with a situation in which the central bank tightens policy more aggressively than markets anticipated in response to improving economic conditions that are publicly observable. In this case, the contractionary effects of tighter monetary policy are more than offset by the expansionary forces associated with the underlying macroeconomic developments.

Turning to the analysis of unconventional MP shocks, our results indicate that CBI effects appear to play a limited role in contaminating the transmission of FG surprises. In contrast, RN shocks tend to distort the estimated effects of these surprises by amplifying and accelerating their impact. When RN shocks are not disentangled, FG innovations appear to generate disproportionately large effects on macroeconomic variables, a pattern that may be consistent with what has been termed the *forward guidance puzzle* in the literature (see Del Negro et al., 2023). Output, for example, declines significantly on impact. In addition, the magnitude of the contraction one month after the shock is roughly 50% larger than the peak decline estimated following a pure target-rate shock, which occurs only after about 15 months. These results are difficult to reconcile with the conventional view that monetary policy affects real activity with substantial lags. They also contrast with the empirical evidence for the U.S. reported by Swanson (2021, 2024), who finds that FG shocks generate macroeconomic effects that are comparable to—or even smaller than—those of conventional MP shocks.

Once RN shocks are isolated, however, the effects of FG innovations become more conservative and broadly similar to those of pure target-rate shocks. One plausible interpretation is that, following episodes in which economic activity fails to cool as expected, the central bank may adopt a more hawkish forward guidance stance to support the convergence of inflation toward target. In such circumstances, hawkish RN shocks may capture not only a tightening implied by the policy rule, but also the economy's ongoing contractionary adjustment already underway. In a VAR that treats raw FG surprises as exogenous shocks, this combination can generate large and immediate contractionary responses, thereby overstating the effects of FG.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 outlines the construction of monetary policy surprises and the identification strategy used to isolate the shocks of interest. Section 4 presents the VAR and the data used in the estimation. The empirical results are reported in Section 5. Section 6 concludes.

1.1 Related literature

This paper relates to the growing body of work that uses HFI to study the macroeconomic effects of monetary policy. Building on early contributions such as Kuttner (2001) and Gürkaynak et al. (2005), a large literature combines interest rate surprises with VARs to estimate the effects of conventional and unconventional MP shocks (e.g., Gertler and Karadi (2015); Nakamura and Steinsson (2018); Andrade and Ferroni (2021); Miranda-Agrippino and Ricco (2023); Bauer and Swanson (2023a, 2023b); Jarociński and Karadi (2020, 2025)).

Our analysis complements Jarociński and Karadi (2025), who jointly identify and measure the effects of CBI and RN components for the U.S., by extending their framework to an EME and by explicitly distinguishing between conventional and unconventional MP shocks. Beyond isolating these components for two types of interest rate surprises, we trace their dynamic effects on a broader set of variables, including long-term bond yields, the exchange rate, and inflation expectations. Notably, we show that RN shocks appear to play distinct roles in the transmission of target-rate and FG shocks.

We also contribute to the literature analyzing the macroeconomic effects of unconventional monetary policy shocks. Campbell et al. (2012) and Andrade and Ferroni (2021) study the macroeconomic effects of forward guidance (FG) in advanced economies, emphasizing that FG surprises may combine "Odyssean" policy commitments with "Delphic" signals about future macroeconomic conditions. Swanson (2024) shows that FG surprises may be contaminated by RN effects. For the case of Mexico, Solís (2023) documents that FG significantly affects the exchange rate and the yield curve, underscoring the importance of forward-looking communication channels in a small open economy.

We add to this literature by providing new evidence on the macroeconomic effects of FG in an EME. We find that unpurged FG surprises generate disproportionately large and faster responses—most notably in real output—resembling what has been described as the FG puzzle. Once RN components are isolated, these distortions are substantially attenuated, and the estimated effects of FG become more comparable to those of conventional MP shocks.

Finally, this paper contributes to the literature on monetary transmission in EMEs, where empirical work using HFI has documented puzzling responses of exchange rates, prices, and inflation expectations to contractionary MP shocks. Aruoba et al. (2021) show that the domestic currency appreciates and inflation expectations rise following an unexpected tightening in Chile. Using a panel of EMEs, Bolhuis et al. (2024) document an average depreciation of the exchange rate after contractionary MP shocks. More recently, Witheridge (2024) reports increases in inflation after an unexpected tightening and interprets them as evidence of fiscal-led policy dynamics in some EMEs, including

Mexico. In contrast, Checo et al. (2024) find more conventional transmission patterns using survey-based forecast errors, while Gomes et al. (2024) document sizable real and financial effects of high-frequency tightening shocks in Brazil. For Mexico, Solís (2023) shows that exchange-rate puzzles largely disappear when monetary policy surprises are identified using intraday data.

We contribute to this literature by proposing an alternative and complementary explanation for the emergence of these puzzles: the failure to account for informational components embedded in monetary policy surprises. We show that once CBI and RN components are isolated, impulse responses become broadly consistent with standard transmission mechanisms. Importantly, we develop the analysis using daily data, which is particularly valuable in EMEs where access to intraday financial data may be limited.

2 High-Frequency Identification of MP, RN, and CBI Shocks

This section outlines the construction of interest rate surprises and the identification strategy used to isolate pure MP, RN, and CBI shocks in Mexico. Our approach closely follows the framework of Jarociński and Karadi (2025) and proceeds in three stages. We begin by constructing target-rate and FG surprises from high-frequency changes in interest rate swaps around monetary policy announcements. In the spirit of Bauer and Swanson (2023a, 2023b), we then regress each policy surprise on a set of macroeconomic and financial variables observable to market participants prior to the announcement. Finally, we combine the fitted and residual components obtained from these regressions, together with equity-price surprises, to identify the aforementioned shocks using sign and magnitude restrictions.

2.1 Constructing Target-Rate and FG Surprises

Following Andrade and Ferroni (2021), Solís (2023a, 2023b), and Bolhuis et al. (2024), we construct monetary policy surprises using high-frequency changes in interest rate swaps around monetary policy announcements. These changes are commonly interpreted as proxying for revisions in market expectations about both the current stance of monetary policy and its expected near-term path.[2] Specifically, we use interest rate swaps with maturities of 1, 3, 6, 9, and 12 months across the short end of the yield curve. Given the lack of intraday data, high-frequency changes in swap rates are measured as close-to-close movements from the trading day preceding the announcement to the day of the announcement. This approach is standard in the literature on EMEs (see Bolhuis et al. (2024), Witheridge (2024), Gomes et al. (2023), and Aruoba et al. (2021)). Our sample includes 214 scheduled monetary policy announcements over the period January 2002 to December 2024. The starting date reflects the availability of bond yield data used in

[2]The Mexican swap market references the 28-day interbank rate (TIIE-28), which is closely tied to the policy rate and serves as a benchmark for short-term bank lending.

the estimations and closely coincides with the adoption of an inflation-targeting regime by Banco de México in 2001.

Following the standard approach of Gürkaynak et al. (2005), we extract the first two principal components (PCs) of the high-frequency changes in swap rates around monetary policy announcements. We then rotate these components so that the second PC has zero loading on the 1-month swap rate. After this rotation, the first factor can be interpreted as reflecting unexpected changes in the current policy rate—closely tracked by the 1-month swap rate—and we therefore label it the *target-rate surprise*. The second factor captures revisions to the expected near-term policy path, that is, movements in longer-maturity swap rates that are orthogonal to changes in the current policy rate, and we refer to it as the *FG surprise*. For comparability and ease of interpretation, both surprises are rescaled so that the target-rate surprise moves one-for-one with changes in the 1-month swap rate, while the FG surprise affects the 12-month swap rate by the same magnitude as the target-rate surprise.

2.2 Decomposing Monetary Policy Surprises into Systematic and Residual Components

In this section, we decompose the target-rate and FG surprises into two components: one that co-moves with publicly available information prior to the announcement, and a residual component orthogonal to this information set. This decomposition follows the logic of Bauer and Swanson (2023a, 2023b), who show that interest rate surprises may reflect not only exogenous policy shocks but also systematic responses of the central bank to economic news that were already observable to market participants, particularly when expectations about the policy reaction function are imperfect.

Formally, for each policy announcement t, we estimate

$$mps_t = \alpha + X'_{t-}\beta + u_t, \tag{1}$$

where mps_t denotes either the target-rate or the FG surprise, X_{t-} collects macroeconomic and financial variables observable to market participants before the announcement (as indicated by the time subscript $t-$), and u_t is the residual. We define the fitted component as $\text{FIT}_t \equiv \widehat{\alpha} + X'_{t-}\widehat{\beta}$ and the residual component as $\text{RES}_t \equiv \widehat{u}_t$, where by construction RES_t is orthogonal to all variables included in X_{t-}. The fitted component captures the portion of the surprise that is explained by the pre-announcement information set and may be interpreted as reflecting revisions in expectations about the systematic policy response to news. The residual component isolates the unpredictable part of the surprise, which is more closely related to deviations from the systematic policy rule.

Empirical studies, including Swanson (2024) and Bauer and Swanson (2023a, 2023b), document a range of macroeconomic and financial variables that help predict monetary policy surprises. We incorporate several of these variables into X_{t-}, while also including a small set of additional variables that may be particularly relevant for an EME such

as Mexico. Given the strong real and financial linkages between the Mexican and U.S. economies, X_{t-} contains both domestic and external variables.[3]

To capture domestic financial conditions, we include the log change in the nominal peso–U.S. dollar exchange rate and changes in the volatility implied by one-month options on the Mexican peso, both measured from three months prior to the announcement to the day before the announcement.[4] In addition, we consider three-month changes in the interbank funding rate and in the two-year Mexican government bond yield.[5] To capture domestic macroeconomic conditions, we include log changes in the Overall Indicator of Economic Activity (IGAE, by its Spanish acronym) and in the core Consumer Price Index (CPI), measured from six months prior to the most recent data release available before the policy announcement.[6,7]

As external financial variables, we include log changes in the Standard & Poor's 500 stock index, changes in two-year and ten-year U.S. Treasury yields, and the log change in Brent crude oil prices, all measured over a three-month horizon prior to the policy announcement. For external macroeconomic conditions, we consider log changes in U.S. industrial production and in the U.S. Consumer Price Index, measured from six months prior to the most recent data release available before the policy announcement. Finally, following Miranda-Agrippino and Ricco (2021) and Swanson (2024), we include one lag of the dependent variable to account for potential serial correlation in monetary policy surprises.

Table 1 reports the results from projecting target-rate and FG surprises onto the set of macroeconomic and financial variables described above, using the full sample from January 2002 to December 2024.[8] Results for target-rate surprises are reported in the first column, while those for FG surprises are shown in the second column. Standard errors

[3]Mexico is a small open economy with strong real and financial linkages to the United States, often viewed as the "center country" in studies of small open economies. Roughly 80% of Mexican exports are directed to the U.S., and a large share of foreign direct investment originates there. Prior work documents close trade and financial integration between the two economies, including substantial co-movement in term premia and short-term interest rates [see, for instance, Fink and Schüler (2015) and Carrillo et al. (2020)].

[4]The volatility implied by one-month options on the Mexican peso reflects market expectations about near-term exchange rate uncertainty. This measure is widely followed by market participants and is regularly discussed in Banco de México's quarterly inflation reports.

[5]The interbank funding rate captures short-term conditions in the domestic money market and is closely tied to the monetary policy rate.

[6]The IGAE follows the methodology and conceptual framework of the national accounts. The correlation between IGAE and GDP, both measured in quarterly percentage changes, is 0.99. The IGAE is subject to revisions, so the data available to market participants and the central bank at a given point in time may differ from the final revised values released by the statistical office. Although the use of real-time data would be desirable, such data are unavailable for Mexican output. We therefore rely on revised data in our estimations.

[7]IGAE is released with a delay of approximately seven to eight weeks (approximated by 42 trading days), while consumer prices are typically released with a lag of about two weeks (approximated by 10 trading days).

[8]Monetary policy surprises, changes in interest rates, and variations in implied exchange rate volatility are expressed in basis points. Changes in real activity, consumer prices, oil prices, asset prices, and the exchange rate are measured as log differences multiplied by 100. Three-month windows correspond to

(in parentheses) are computed using heteroskedasticity- and autocorrelation-consistent (HAC) estimators. As can be seen, the set of regressors explains a non-negligible share of the variation in monetary policy surprises, with R^2 values of approximately 22% for target-rate surprises and 15% for FG surprises, broadly in line with estimates reported in the related literature.[9,10]

These results indicate that monetary policy surprises appear to contain a systematic component that co-moves with publicly available economic information observed prior to policy announcements. While the underlying economic mechanisms driving this relationship are beyond the scope of this exercise, its presence is consistent with the possibility that market expectations about the systematic policy response are imperfect. For the VAR analysis, however, the precise interpretation of this correlation is of secondary importance, as emphasized by Swanson (2024). What matters is the empirical regularity that monetary policy surprises are not fully orthogonal to pre-announcement economic and financial conditions. As we show below, failing to account for this feature leads to impulse response functions that display puzzling or potentially misleading dynamics in response to an MP shock.

In the next section, we exploit the joint high-frequency behavior of the fitted and residual components of monetary policy surprises together with equity-price surprises to identify pure MP, RN, and CBI shocks through sign and magnitude restrictions.

2.3 Identifying MP, RN, and CBI Shocks with Sign and Magnitude Restrictions

Following Jarociński and Karadi (2025), we combine the previously obtained fitted and residual components with equity-price surprises to identify pure MP, RN, and CBI shocks using sign and magnitude restrictions. The identification is implemented separately for target-rate and FG surprises. In each case, we use the corresponding fitted and residual components—denoted by $(\mathrm{FIT}^{TR}, \mathrm{RES}^{TR})$ for target-rate surprises and $(\mathrm{FIT}^{FG}, \mathrm{RES}^{FG})$ for FG surprises—together with a common equity-price surprise measure, SP. The latter is constructed as the log difference of the Mexican Stock Market Index (IPC) between the trading day preceding the policy announcement and the day of the announcement.

65 trading days. U.S. consumer prices and U.S. industrial production are typically released with a lag of about two weeks.

[9]We also estimate LASSO-based projections as a complementary diagnostic and robustness exercise (see Table AX in the Appendix). For target-rate surprises, the predictors selected by LASSO broadly align with those included in the baseline information set, whereas for FG surprises the regularization procedure selects a substantially smaller subset of variables. As shown in Figures AX and AY in the Appendix, impulse responses based on LASSO-selected controls remain broadly consistent with our baseline results but exhibit a weaker separation between pure MP and RN shocks, making the identification less sharp. For this reason, we do not adopt the LASSO-based specification as our baseline empirical model.

[10]We also consider an extended specification that augments the projection with a broader set of macro-financial controls. The fitted and residual components remain largely unchanged, and the resulting VAR impulse responses are very similar to those obtained under the baseline specification (see Table BX and Figures BX and BY in the Appendix).

Our identification strategy assumes that, within a daily window around policy announcements, interest rate and equity-price surprises are predominantly driven by three structural disturbances—pure MP, RN, and CBI shocks.[11] In addition, we assume that each disturbance generates a distinct pattern of co-movement between interest rate and equity-price surprises.

A ***pure MP shock*** is characterized by a negative co-movement between the residual component of the interest rate surprise—either RES^{TR} or RES^{FG}—and the stock price surprise. This pattern is consistent with an unexpected policy tightening that raises discount rates and lowers expected future cash flows, leading to an immediate decline in equity prices. When identification is based on target-rate surprises, we refer to this disturbance as a *pure target-rate shock*; when based on FG surprises, as a *pure FG shock*.

An ***RN shock*** is identified by a negative co-movement between the fitted component of the interest rate surprise—either FIT^{TR} or FIT^{FG}—and the stock price surprise. This shock captures episodes in which the central bank responds more aggressively than markets anticipated to publicly available information that was already incorporated into expectations prior to the announcement. As a result, the main information revealed within the announcement window is the stronger-than-expected monetary policy response, which depresses equity prices through standard discount-rate and cash-flow channels.

A ***CBI shock*** is identified by a positive co-movement between the overall monetary policy surprise—defined as $\text{FIT}^j + \text{RES}^j$—and the equity-price surprise. This pattern may arise when the central bank reveals favorable new information about economic fundamentals while simultaneously tightening policy. In such cases, equity prices increase in response to improved growth prospects and higher expected future cash flows, despite the policy tightening.

These assumptions are summarized by the following impact representation:

$$\begin{pmatrix} \text{RES}^j \\ \text{FIT}^j \\ \text{SP} \end{pmatrix} = \begin{pmatrix} + & c_{12} & c_{13} \\ c_{21} & + & c_{23} \\ - & - & + \end{pmatrix} \begin{pmatrix} \text{MP}^j \\ \text{RN}^j \\ \text{CBI}^j \end{pmatrix}, \quad j \in \{\text{TR}, \text{FG}\}, \tag{2}$$

together with the restriction that the overall policy surprise (the sum of fitted and

[11] This assumption is stronger than in studies for AEs that rely on narrow intraday windows surrounding policy announcements. Nevertheless, intraday financial data are typically unavailable for EMEs, and much of the literature therefore relies on daily windows for this type of analysis. One exception is Solís (2023a, 2023b), who employs intraday data on interest rate swaps for Mexico. However, that dataset covers a shorter sample period starting in 2011 and does not include intraday equity-price data, which are central to our identification strategy. As emphasized by Solís (2023a, 2023b), measurement error in daily swap-rate changes appears to be limited, as the correlation between intraday and daily changes in swap rates is close to one. To further alleviate concerns that the equity-price surprise may partly reflect global financial developments unrelated to domestic monetary policy news, we orthogonalize our equity-price surprise with respect to contemporaneous changes in the S&P 500 index as a robustness check. The results obtained using this orthogonalized measure are very similar to those reported in the baseline specification and are presented in Figure BW in the Appendix.

residual components) rises on impact after a favorable CBI shock:

$$(c_{13} + c_{23}) > 0. \tag{3}$$

Finally, we impose magnitude restrictions to ensure that the FIT–RES decomposition provides informative proxies for the MP and RN shocks:

$$|c_{12}| < c_{11}, \tag{4}$$

$$|c_{21}| < c_{22}. \tag{5}$$

Restriction (4) ensures that the residual component loads more strongly on the MP shock than on the RN shock, while restriction (5) ensures that the fitted component loads more strongly on the RN shock than on the MP shock.

3 VAR Model for a Small Open Economy

To analyze the macroeconomic effects of MP, RN, and CBI shocks, we estimate a monthly Bayesian VAR model. The reduced-form representation of the VAR is given by

$$\boldsymbol{Y}_t = \boldsymbol{c} + \sum_{p=1}^{P} \boldsymbol{B}^p \boldsymbol{Y}_{t-p} + \boldsymbol{\delta}\, \boldsymbol{Z}_t + \boldsymbol{u}_t, \tag{6}$$

where the vector $\boldsymbol{Y}_t$ stacks high-frequency surprises and low-frequency macroeconomic and financial variables observed in month t, $\boldsymbol{c}$ collects intercepts, $\boldsymbol{B}^p$ denotes the lag-p coefficient matrices, $\boldsymbol{Z}_t$ contains exogenous foreign variables capturing international conditions, and $\boldsymbol{u}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma})$. In particular, we partition $\boldsymbol{Y}_t$ as

$$\boldsymbol{Y}_t = \begin{pmatrix} \boldsymbol{m}_t^j \\ \boldsymbol{y}_t \end{pmatrix}, \qquad j \in \{\mathrm{TR}, \mathrm{FG}\},$$

where $\boldsymbol{m}_t^j$ collects the fitted and residual components $(\mathrm{FIT}_t^j, \mathrm{RES}_t^j)$ associated with either the target-rate or the FG surprise, together with the equity-price surprise SP_t. In turn, $\boldsymbol{y}_t$ denotes a vector of domestic macroeconomic and financial variables described below. As in Jarociński and Karadi (2020, 2025) and Breitenlechner et al. (2021), the high-frequency surprises are aggregated to the monthly frequency by summing surprises within month t, assigning a value of zero to months with no policy announcements. Following previous studies for small open economies, we include foreign exogenous controls $\boldsymbol{Z}_t$—primarily U.S. variables—which are allowed to affect domestic variables contemporaneously, but not vice versa.[12] In our baseline specification, we include one lag, which is common in studies of monetary policy transmission in EMEs.[13] The set

[12]See, for instance, Bjørnland (2009), Ibarra (2016), Capistrán et al. (2019), and Kim and Lim (2022), among others.
[13]See, e.g., Canova (2005), Georgiadis (2016), and Dahlhaus et al. (2018).

of domestic macroeconomic and financial variables, as well as the exogenous foreign variables included in the VAR model, is presented in Table 3.[14]

Regarding Mexican macro-financial variables, we include the one-year government bond yield, i_t^{1y}, and the ten-year government bond yield, i_t^{10y}.[15] The one-year yield reflects both the current policy stance and expectations about near-term policy adjustments (Gertler and Karadi, 2015). The ten-year yield allows us to assess the impact on long-term rates, which are especially relevant for firms' and households' decisions (Andrade and Ferroni, 2021). For real activity, we use an interpolated monthly GDP series, y_t, using the approach of Elizondo (2012).[16] For prices, we use the core Consumer Price Index, p_t, which is seasonally adjusted with the X13-ARIMA method. We also include the nominal peso–U.S. dollar exchange rate, s_t, the Mexican Stock Market Index, ipc_t, and 12-month-ahead inflation expectations, denoted by π_t^e.

In the vector of exogenous controls $\boldsymbol{Z}_t$, we include the U.S. two-year Treasury yield, i_t^{2*}, as movements in U.S. rates spill over to Mexican yields via portfolio rebalancing and term-structure arbitrage, thereby affecting domestic macroeconomic and financial conditions. As discussed by Swanson and Williams (2014), Gertler and Karadi (2015), and Bauer and Swanson (2023a), the two-year Treasury yield was essentially unconstrained during the zero lower bound period (2009–2015), making it an informative measure of the overall stance of U.S. monetary policy. We also include the S&P 500 index to capture shifts in external financial conditions that may propagate to Mexico. U.S. industrial production and the consumer price index, (y_t^*, p_t^*), proxy foreign demand and external price pressures, respectively. The VIX index is included as a standard proxy for risk aversion (Gambacorta et al., 2014); (MacDonald and Popiel, 2020), and the Brent crude oil price, given its relevance for Mexico's terms of trade (Capistrán et al., 2019). Finally, following Ng (2021), Evgenidis and Fasianos (2023), and Carrillo et al. (2025), among others, we include monthly dummy variables to account for the unprecedented disruptions associated with the COVID-19 pandemic in our estimates. Specifically, we include separate dummies for each month from April to September 2020, which allow for temporary intercept shifts during the most disrupted period.[17]

We estimate the model using a Bayesian approach, which is standard in the sign restrictions literature. In particular, we employ the Minnesota prior of Litterman (1979), with an overall tightness parameter set to 0.2. Further details are provided in

[14] All domestic and foreign financial variables are measured at end-of-month values.

[15] We use zero-coupon yields at both maturities to ensure comparability, since coupon-bearing bonds differ across maturities.

[16] Elizondo (2012) employs the Kalman filter to interpolate the monthly GDP series, utilizing the Overall Indicator of Economic Activity (IGAE by its Spanish acronym) available at a monthly frequency. As a robustness exercise, we use the IGAE index as an alternative measure of economic activity in Mexico. The results, reported in Figures B.5 and B.6 of Appendix B, are consistent with our baseline specification.

[17] April–September 2020 corresponds to the period of particularly severe economic disruptions in Mexico (Banco de México, 2020). A related approach commonly used in the literature is to construct outlier dummies by flagging observations that exceed n times the interquartile range from the sample median (Stock and Watson, 2002). Setting $n = 2.5$ and examining outliers in measures of economic activity yields a set of pandemic dummies that is essentially unchanged.

Appendix A. Posterior inference is conducted using Gibbs sampling techniques, which allow us to approximate the marginal posterior distributions of the model parameters by drawing from their conditional distributions. As in Jarociński and Karadi (2020, 2025) and Breitenlechner et al. (2021), we report results based on 2,000 posterior draws. Specifically, the estimation is implemented using 10,000 iterations of the Gibbs sampler, of which the first 2,000 are discarded as burn-in, and every fourth iteration is retained thereafter.[18]

For structural inference, we follow Rubio-Ramírez, Waggoner, and Zha (2010) and draw structural shocks and impulse responses using a uniform prior over admissible orthogonal rotations, conditional on the imposed sign and magnitude restrictions described above. Because these restrictions deliver *set identification*, each posterior draw of the reduced-form parameters may admit multiple orthogonal rotations consistent with the identifying assumptions. When constructing uncertainty bands, we therefore integrate over both sources of uncertainty—reduced-form parameter uncertainty and rotational uncertainty—by assigning equal prior weight to all admissible rotations.[19]

Finally, we also analyze the macroeconomic effects of an MP shock without disentangling the RN and CBI components. To this end, we estimate the VAR using the overall monetary policy surprise—either the target-rate or the FG surprise—together with the same set of low-frequency macroeconomic and financial variables included in $\boldsymbol{y}_t$ and the exogenous controls in $\boldsymbol{Z}_t$. In this specification, the MP shock is identified by applying a Cholesky decomposition to the covariance matrix of reduced-form innovations, ordering the monetary policy surprise first. This ordering implies that all macroeconomic and financial variables are allowed to respond contemporaneously to the shock, consistent with the standard HFI approach. This identification strategy has been widely used in the literature, including by Jarociński and Karadi (2020), Plagborg-Møller and Wolf (2021), and Bianchi et al. (2022), among others.

[18]We verify that extending the chain length by a factor of ten yields essentially identical results.

[19]To draw structural shocks and impulse responses, we exploit the assumption that high-frequency surprises are affected only by MP, RN, and CBI shocks within the announcement window, and not by other structural disturbances. This timing assumption implies a block-recursive structure for the reduced-form innovations, with the three high-frequency announcement variables ordered first and forming the initial block of shocks. Conditional on this ordering, we impose the sign and magnitude restrictions on the contemporaneous responses to the first three shocks following Rubio-Ramírez, Waggoner, and Zha (2010). For each posterior draw of the reduced-form parameters (B, Σ), we compute the lower-triangular Cholesky factor C of Σ and search over orthogonal rotations that act only on the first three shocks. Specifically, let $Q = \mathrm{diag}(Q^*, I_{n-3})$, where $Q^* \in \mathbb{R}^{3\times 3}$ is obtained from the Q factor in the QR decomposition of a 3×3 matrix with i.i.d. $\mathcal{N}(0, 1)$ entries. For each candidate rotation Q, we form $A_0 = CQ$ and verify whether the resulting impact responses satisfy the imposed sign and magnitude restrictions. All rotations that meet these restrictions are retained and assigned equal (uniform/Haar) prior weight within the admissible set. Each accepted A_0 provides a draw of contemporaneous responses, from which dynamic impulse responses are computed using the VAR coefficients. As emphasized by Jarociński and Karadi (2020), a uniform prior over rotations is less restrictive than enforcing sign restrictions via penalty functions, as in Uhlig (2005).

4 The Effects of MP, RN, and CBI Shocks in Mexico

4.1 The Macroeconomic Effects of Target-Rate Shocks

This section analyzes the macroeconomic effects of target-rate shocks, distinguishing between pure MP, RN, and CBI components. We proceed in two steps. First, we report impulse responses obtained under the *standard HFI* approach, which treats the raw target-rate surprise as a single MP shock and does not separately identify RN and CBI shocks. Second, we present results based on the sign- and magnitude- restriction identification strategy, which disentangles pure target-rate shocks from RN and CBI shocks. All shocks are normalized to correspond to a one percentage point increase in the one-year rate following an MP disturbance.[20] We report median responses over a 72-month horizon, together with 68% highest posterior density intervals (HPDIs) corresponding to the 16th and 84th percentiles of the posterior distribution.[21]

Panel A of Figure 1 presents impulse responses to a contractionary target-rate shock *without* disentangling the RN and CBI components. As can be seen, these shocks are initially followed by an expansion in output, rising prices, a depreciation of the domestic currency (i.e., an increase in the exchange rate), and higher inflation expectations. Although output and prices eventually decline, the responses are considerably attenuated and reach their trough with a substantial lag. The peak contraction in output is approximately 0.27% and occurs around two years after the shock, while the peak decline in prices is about 0.11% and occurs more than four years after the disturbance. A similar pattern emerges for the Mexican peso, which exhibits pronounced delayed overshooting, reaching its maximum appreciation more than two years after the shock, with a relatively modest magnitude of approximately 0.97%. These dynamics stand in contrast to the predictions of standard macroeconomic models and may indicate that the raw target-rate surprises are contaminated by CBI and/or RN components. It is also worth noting that, at first glance, the responses of stock prices and bond yields do not appear to exhibit a clear contamination bias. As we show below, however, they do.

Panel B of Figure 1 presents impulse responses obtained after disentangling pure MP, RN, and CBI shocks from target-rate surprises. The first column reports impulse responses to a contractionary target-rate shock that is now purged of RN and CBI components. Notably, once these components are isolated, the previously documented output, price, and exchange-rate puzzles largely disappear. In addition, inflation expectations exhibit a negative, albeit muted, response, suggesting that they remain relatively well anchored following the unexpected tightening. Compared to the results

[20] Standardizing the size of the shocks facilitates comparability with the existing literature, where a one percentage point increase in interest rates is commonly used as a benchmark. As is typical in the HFI literature, the estimated monetary policy surprises are small relative to raw changes in interest rates, reflecting the fact that financial markets anticipate most policy rate decisions (Nakamura and Steinsson, 2018). In Figure B.15 of Appendix B, we report impulse responses without normalization.

[21] The use of 68% confidence intervals is standard in closely related studies (see, among others, Eichenbaum and Evans, 1995; Kim and Roubini, 2000; Kim, 2001; Faust and Rogers, 2003; Scholl and Uhlig, 2008; Rogers et al., 2018; Breitenlechner et al., 2021) and in the VAR literature more broadly (e.g., Sims and Zha, 1999).

in Panel A, the contraction in output is substantially stronger and materializes earlier, with the peak decline occurring nine months sooner and reaching around 0.41% below the baseline. A similar pattern emerges for prices, whose peak contraction is more than twice as pronounced (around 0.28%) and occurs roughly two years after the disturbance—approximately half the time required in Panel A. The exchange-rate response also changes markedly. Instead of the small and delayed appreciation observed following the raw disturbance, the Mexican peso now appreciates by approximately 4.45% one month after the shock—a magnitude more than four times larger—before gradually returning toward its pre-shock level. Stock prices display a much stronger impact response as well, declining by about 4.27% compared to roughly 1.6% after the unpurged shock. Finally, although both the one-year and ten-year government bond yields increase on impact, their responses appear less persistent than in Panel A, suggesting that the tightening implied by the pure shock is shorter-lived and more consistent with a temporary policy adjustment.

We now describe the impulse responses to hawkish RN shocks and favorable CBI shocks reported in the second and third columns of Panel B of Figure 1, respectively. This analysis helps clarify the role of each component in contaminating the transmission of target-rate shocks. In line with Jarociński and Karadi (2020, 2025), favorable CBI shocks are followed by a persistent increase in the one-year government bond yield, an expansion in output, rising prices, a depreciation of the domestic currency, and an increase in inflation expectations. Stock prices also rise on impact, while the ten-year government bond yield exhibits a positive, albeit muted, response. In general, these responses closely resemble the puzzling dynamics observed following raw target-rate shocks, indicating that CBI effects account for a substantial share of the apparent anomalies.

In particular, the depreciation of the domestic currency in response to favorable CBI shocks may be consistent with upward revisions in inflation expectations, as emphasized by Gürkaynak et al. (2021) and Stavrakeva and Tang (2019). This result is noteworthy, as there remains active debate regarding the exchange-rate response to CBI shocks.[22] An additional mechanism that may help explain this result relates to Mexico's tight integration with the United States. If the favorable news embedded in the CBI shock primarily reflects a stronger U.S. outlook accompanied by dollar appreciation, global investors may rebalance portfolios toward U.S. assets, leading to a weaker peso. At the same time, the muted response of the ten-year government bond yield following a favorable CBI shock may reflect an offsetting mechanism. The stimulative effects of favorable information about economic fundamentals may reduce the term premium,

[22]While our findings are broadly consistent with U.S. evidence (see Pinchetti and Szczepaniak, 2023; Gründler et al., 2022; Stavrakeva and Tang, 2019), evidence for the euro area points in the opposite direction. Kerssenfischer (2022) finds that CBI shocks are followed by an appreciation of the domestic currency, potentially reflecting yield-seeking capital reallocation: higher domestic interest rates and stronger fundamentals may attract capital inflows (Koepke, 2019), thereby strengthening the domestic currency.

thereby counteracting the upward pressure on long-term rates induced by the policy tightening.[23]

Taken together, these responses are consistent with an interpretation in which the central bank reveals favorable information about economic fundamentals while simultaneously tightening policy to lean against emerging overheating and inflationary pressures. Such pressures may operate through an aggregate demand channel and/or through exchange-rate pass-through effects. The persistent increase in the one-year government bond yield is consistent with a systematic policy response aimed at containing these pressures. While the tightening does not fully offset the initial expansionary effects of the news, it gradually neutralizes them over the medium term.

We now describe the impulse responses to hawkish RN shocks reported in the second column of Panel B of Figure 1. These shocks are followed by persistent increases in both short- and long-term interest rates, an expansion in output, rising prices, a depreciation of the domestic currency, and higher inflation expectations. These responses are consistent with Bauer and Swanson (2023a, 2023b) and closely resemble the dynamics observed following raw target-rate shocks, indicating that RN effects account for a substantial share of the apparent anomalies associated with those shocks. Notably, hawkish RN shocks also appear to explain the persistent increase in the ten-year government bond yield observed after a raw target-rate shock.

Overall, these responses appear to be consistent with a scenario in which the central bank tightens policy more aggressively than markets had anticipated in response to improving economic conditions or perceived upside risks to inflation. In this case, the persistent increase in the one-year government bond yield reflects a systematic policy reaction to publicly observable developments. As in the CBI case, the tightening does not fully offset the initial expansionary effects associated with the underlying fundamentals, but gradually neutralizes them over the medium term. These results align with the conceptual framework of Jarociński and Karadi (2025), in which impulse responses to hawkish RN shocks reflect a combination of the underlying economic developments that trigger the central bank's reaction and the unexpected tightening itself.

4.2 The Macroeconomic Effects of FG Shocks

This section analyzes the macroeconomic transmission of news about the *future* path of monetary policy, focusing on FG shocks. While the effects of FG have been extensively studied for AEs, systematic evidence for EMEs remains limited. To the best of our knowledge, this section provides some of the first evidence on the macroeconomic effects of FG shocks in an EME by explicitly disentangling RN and CBI components. As in the previous section, we proceed in two steps. First, we discuss impulse responses obtained under the *standard HFI* specification, which treats the FG surprise as a single MP shock

[23]The term premium is the additional compensation that investors require for holding longer-term instruments instead of rolling over short-term debt. According to Carrillo et al. (2020), the response of the term premium depends on whether the information conveyed by the announcement is perceived as negative or positive for the economic outlook. The term premium tends to rise when news signals a deterioration in fundamentals and to fall when news points to a more favorable outlook.

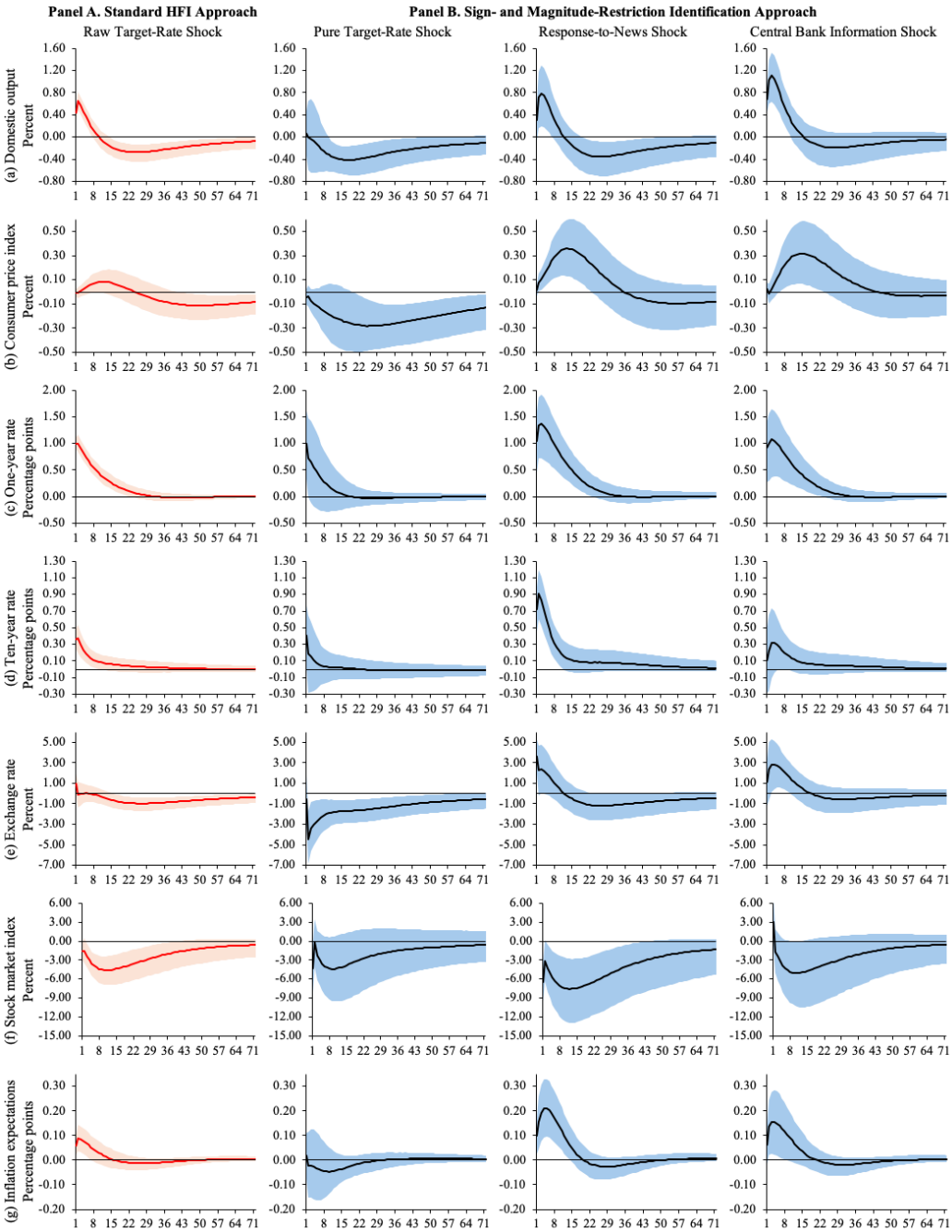


Figure 1 Impulse Responses to Target-Rate, Response-to-News, and Central Bank Information Shocks

Notes: Responses are presented for a 72-month horizon with the associated 68% highest posterior density intervals (HPDIs).

and does not disentangle RN and CBI components. Second, we present results based on the sign- and magnitude-restriction identification, which separately identifies pure FG, RN, and CBI shocks. For comparison, we also reproduce the impulse responses to a contractionary pure target-rate shock reported in Panel B of Figure 1. All shocks are normalized to correspond to a one percentage point increase in the one-year rate following an MP shock.

Panel A of Figure 2 displays, in red, impulse responses to a contractionary FG shock identified under the standard HFI approach, that is, without disentangling RN and CBI components. Impulse responses to a contractionary pure target-rate innovation are reported in green. The plots convey two main messages. First, in contrast to the results for conventional MP shocks presented previously, unpurged FG innovations do not exhibit a clear attenuation bias arising from RN and CBI components. The responses to a contractionary FG shock broadly display the conventional qualitative signs associated with monetary tightening: both the one-year and ten-year yields increase, output contracts, prices decline, equity prices fall, the Mexican peso appreciates, and inflation expectations remain anchored.

However, compared to the effects of pure target-rate shocks, unpurged FG shocks generate disproportionately large and faster effects on domestic output—a pattern consistent with what has been termed the *forward guidance puzzle* in the literature (see Del Negro et al., 2023). Output declines by around 0.44% one month after the shock, exceeding the maximum contraction observed following a target-rate shock, which materializes only after approximately 19 months. These dynamics are difficult to reconcile with the conventional view that monetary policy affects real activity with substantial lags, as well as with the U.S. evidence reported by Swanson (2021, 2024), who finds that FG shocks generate effects that are comparable to—or even smaller than—those of conventional MP shocks. Moreover, the responses of prices and financial variables—such as the exchange rate and stock prices—also suggest an overstatement of the true effects of FG shocks.

Panel B of Figure 2 reports impulse responses to the different structural components extracted from FG surprises. As before, we also plot in green the impulse responses to a contractionary pure target-rate shock. Several results emerge from this analysis. First, as shown in the first column, once RN and CBI components are isolated, the contractionary effects of FG shocks become considerably more conservative and closely resemble those of pure target-rate disturbances. This is reflected in the considerable overlap between their impulse responses across macroeconomic and financial variables. Notably, the large and immediate decline in output estimated under the standard HFI approach largely disappears. These findings are more consistent with standard monetary transmission mechanisms and with the recent international evidence reported in Swanson (2021, 2024).

Second, in contrast to the results for conventional MP shocks discussed in the previous section, favorable CBI innovations extracted from FG surprises do not systematically exhibit expansionary effects, thereby implying a limited attenuation bias in the estimated effects of FG shocks. One possible interpretation is that the central bank responds to favorable information about economic fundamentals primarily by adjusting the current policy rate, rather than by modifying its communication about the future path of

monetary policy. Consequently, CBI shocks may be more relevant for contaminating target-rate surprises than FG surprises.

Third, in contrast to hawkish RN shocks identified from target-rate surprises—which are expansionary and tend to attenuate or even reverse the contractionary effects of conventional MP shocks—hawkish RN shocks identified from FG surprises are contractionary and instead distort the estimated effects of FG by amplifying and accelerating their impact. One plausible interpretation is that, following episodes in which economic activity fails to cool as expected, the central bank adopts a more hawkish FG stance to signal a stronger or more persistent commitment to bringing inflation back to target. In such circumstances, RN shocks capture not only the unexpected tightening implied by the policy rule, but also the economy's ongoing contractionary adjustment already underway. In a VAR that treats raw FG surprises as exogenous shocks, this combination can generate large and immediate contractionary responses, thereby overstating the true effects of FG.

4.3 The Importance of Jointly Controlling for RN and CBI

In this exercise, we compare impulse responses to a contractionary MP shock obtained under four alternative identification strategies. First, the *standard HFI* approach treats the raw monetary policy surprise as a single MP shock and does not separately identify RN or CBI components. Second, the Jarociński and Karadi (2020) approach controls for CBI shocks but does not isolate RN components. Third, the Bauer and Swanson (2023a, 2023b) approach controls for RN shocks but does not account for CBI effects. Finally, the Jarociński and Karadi (2025) framework jointly controls for both RN and CBI shocks. All shocks are normalized to correspond to a one percentage point increase in the one-year government bond yield to ensure comparability across specifications.

Figure 3 presents the corresponding impulse responses to a contractionary target-rate shock. As can be seen, controlling for only one informational channel, either RN or CBI, helps mitigate the anomalies observed under the standard HFI approach. However, the impulse responses remain attenuated and, in some cases, continue to display counterintuitive dynamics. Only when both RN and CBI components are jointly isolated do the impulse responses become more closely aligned with the predictions of standard macroeconomic models. In this case, output and prices decline more sharply and with shorter delays, the domestic currency appreciates more strongly, and inflation expectations exhibit a clearer negative median response.

Figure 4 presents the corresponding impulse responses to a contractionary FG shock. As documented in the previous section, the main issue with these shocks under the standard HFI approach is that they generate disproportionately large and faster effects—most notably on domestic output. Controlling for CBI shocks alone does not mitigate these anomalies. On the contrary, it appears to amplify them, as evidenced by the stronger responses of output, prices, and the exchange rate. By contrast, controlling for RN shocks substantially attenuates these distortions: the contractionary effects of FG shocks become more moderate and closely resemble those identified under the Jarociński and Karadi (2025) framework. A remaining issue, however, is that inflation expectations

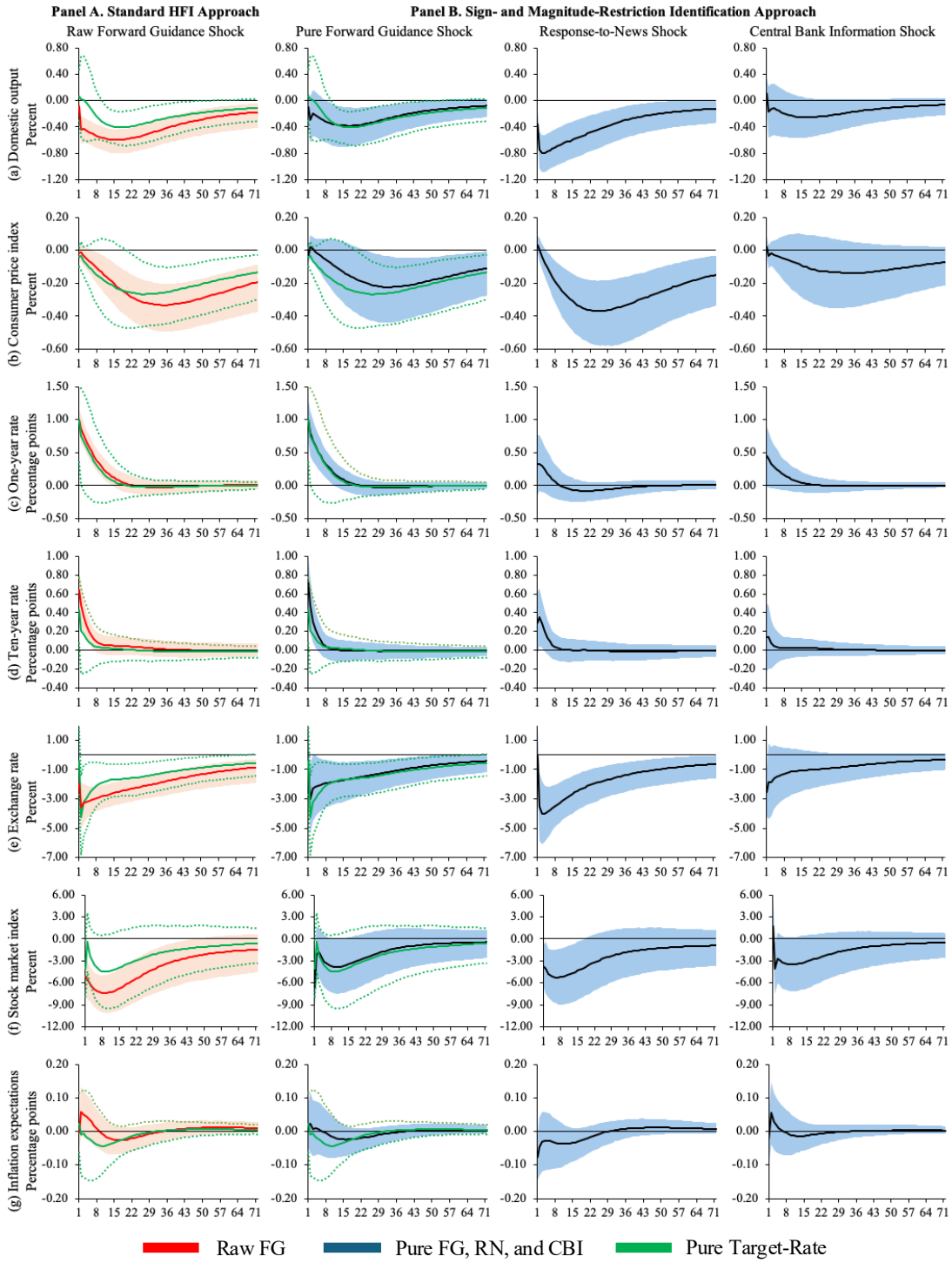


Figure 2 Impulse Responses to Forward Guidance, Response-to-News, and Central Bank Information Shocks

Notes: Responses are presented for a 72-month horizon with the associated 68% highest posterior density intervals (HPDIs).

exhibit a puzzling increase in response to the contractionary shock when only RN shocks are controlled for. This anomaly is largely resolved once both RN and CBI components are jointly isolated.

Taken together, these results suggest that partial purification of monetary policy surprises may be insufficient. Failing to account simultaneously for both informational channels can lead to biased impulse responses, thereby distorting inference about the true transmission of monetary policy.

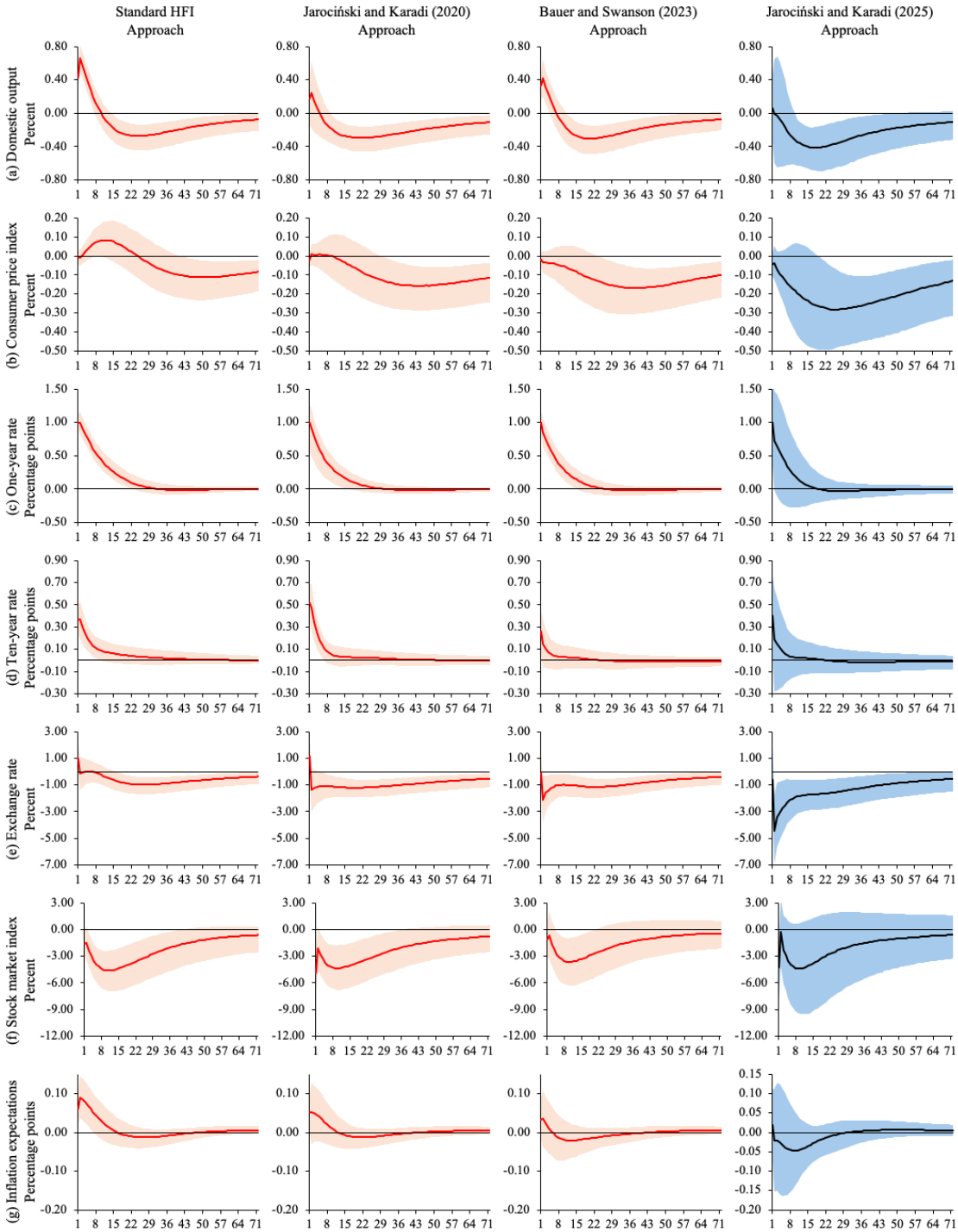


Figure 3 Impulse Responses to Target-Rate Shocks under different Approaches

Notes: Responses are presented for a 72-month horizon with the associated 68% highest posterior density intervals (HPDIs).

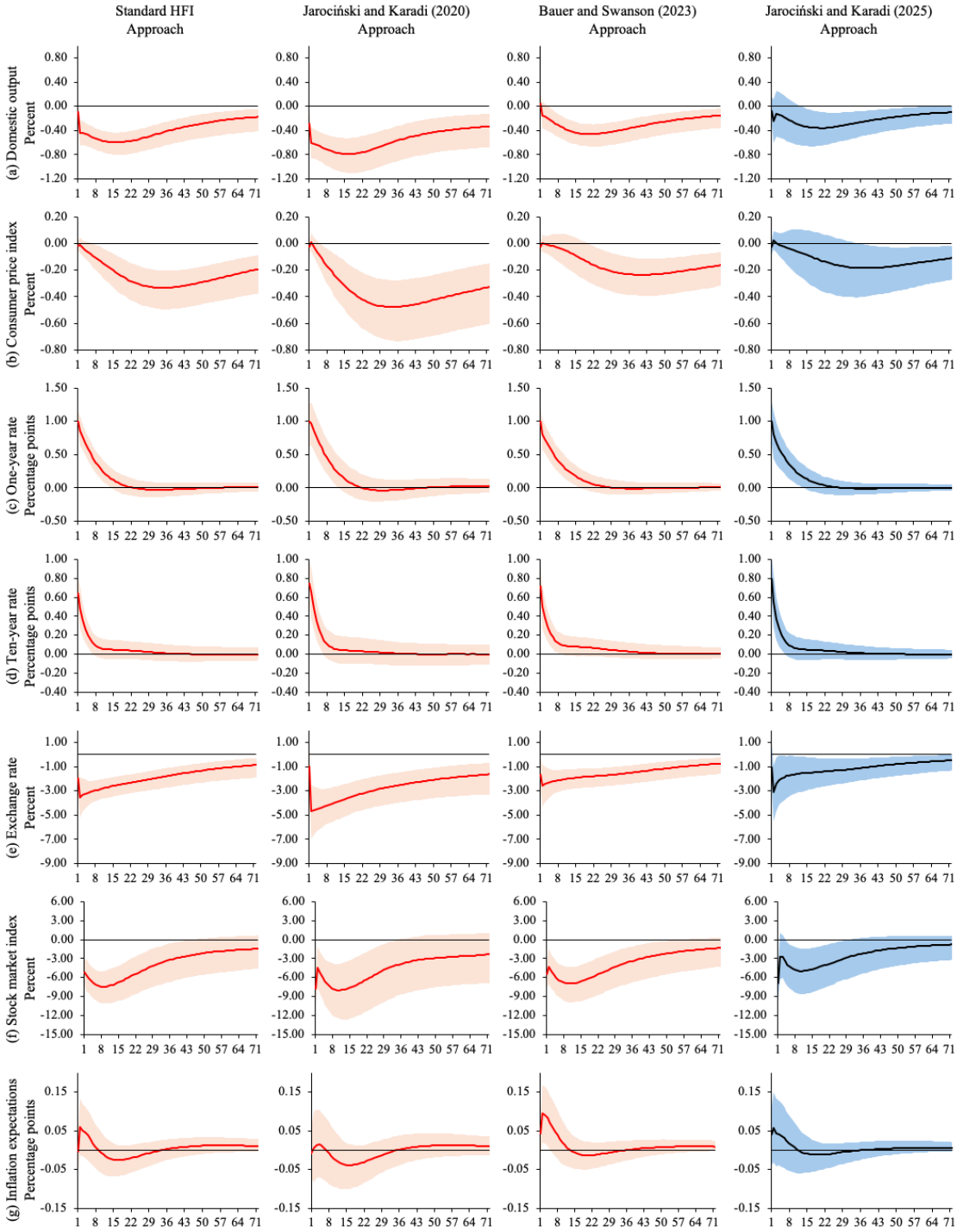


Figure 4 Impulse Responses to Forward Guidance Shocks under different Approaches

Notes: Responses are presented for a 72-month horizon with the associated 68% highest posterior density intervals (HPDIs).

Table 1 Regressions of High-Frequency Monetary Policy Surprises on Macroeconomic and Financial News

	Target-rate surprise	FG surprise
Domestic news		
Δ log domestic output (6m)	−0.036	0.239
	(0.177)	(0.674)
Δ log CPI (6m)	−0.287	2.109
	(1.246)	(3.483)
Δ log MXN/USD (3m)	0.007	0.962**
	(0.168)	(0.410)
Δ FX implied volatility (3m)	0.004*	−0.010**
	(0.002)	(0.004)
Δ Interbank rate (3m)	0.022	−0.010
	(0.015)	(0.027)
Δ 2-year gov. yield (3m)	0.030*	0.008
	(0.015)	(0.039)
External news		
Δ log Industrial prod. (6m)	0.007	0.782
	(0.261)	(0.849)
Δ log US CPI (6m)	1.201	−1.219
	(0.979)	(2.053)
Δ log S&P 500 (3m)	0.129	0.252
	(0.143)	(0.337)
Δ 2-year Treasury yield (3m)	−0.007	0.091**
	(0.021)	(0.040)
Δ 10-year Treasury yield (3m)	−0.052*	−0.095*
	(0.026)	(0.054)
Δ log Brent oil price (3m)	0.078*	0.123
	(0.045)	(0.110)
Lagged monetary policy surprises		
Surprise_{t-1}	0.021	−0.225**
	(0.120)	(0.097)
R^2	0.224	0.154
Adjusted R^2	0.173	0.099
Observations	213	213

Notes: This table reports coefficient estimates from regressions of target-rate and forward guidance surprises on macroeconomic and financial news released prior to the monetary policy announcement. Standard errors (in parentheses) are heteroskedasticity- and autocorrelation-consistent (HAC), computed using a Newey–West estimator. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Series	Definition	Transformation	Source
	Domestic variables in $\boldsymbol{y}_t$		
i_t^{1y}	1-year government bond yield	No transformation	Banco de México
i_t^{10y}	10-year government bond yield	No transformation	Banco de México
ipc_t	Mexican Stock Market Index	$\ln(ipc_t)$	Grupo BMV
s_t	MXN per USD exchange rate	$\ln(s_t)$	Banco de México
y_t	Interpolated GDP	$\ln(y_t)$	Elizondo (2012)
p_t	Core consumer price index (CPI)	$\ln(p_t)$	INEGI
π_t^e	12-month-ahead inflation expectations	No transformation	Banco de México
	Exogenous variables in $\boldsymbol{Z}_t$		
i_t^{2*}	U.S. 2-year Treasury yield	No transformation	FRED
$sp500_t$	S&P 500 index	$\ln(sp500_t)$	Bloomberg
vix_t	VIX index	$\ln(vix_t)$	Bloomberg
y_t^*	U.S. Industrial Production	$\ln(y_t^*)$	FRED
p_t^*	U.S. CPI	$\ln(p_t^*)$	FRED
oil_t	Brent crude oil price	$\ln(oil_t)$	FRED

Table 2 Data Description

5 Robustness in a HANK and SAM Model

The main results of this paper are presented in a partial equilibrium setup with aggregate demand effects that do not arise from general equilibrium effects. We think there are many advantages to studying the welfare and multiplier effects in this setting without embedding the model in general equilibrium. First, general equilibrium models often struggle to adequately capture the feedback mechanisms between consumption and income, particularly the asymmetric nature of these relationships during recessionary versus expansionary periods. Additionally, a complete general equilibrium treatment would necessitate the analysis of numerous complex channels including investment dynamics, firm ownership structures and dividend distribution policies, inventory management, and international trade flows—elements that, while important in their own right, would potentially obscure the core mechanisms we aim to investigate.

Despite the advantages of our partial equilibrium approach, here we complement our analysis with a general equilibrium HANK and SAM model, as standard as possible, that is able to capture supply-side effects that are absent from the partial equilibrium model. In particular, fiscal policies can generate labor market responses that our partial equilibrium analysis does not address. These supply-side channels can affect both the welfare implications and the fiscal multipliers of different policy interventions. Furthermore, standard to the HANK and SAM literature, the general equilibrium model generates a self-reinforcing precautionary saving channel that amplifies business cycles. During a recession, heightened unemployment risk prompts households to increase savings and reduce consumption which in turn weakens both aggregate and labor demand. The resulting decline in labor demand further raises unemployment risk, reinforcing precautionary savings.

We embed the consumption choices of our households——with heterogeneity over education type and discount factors——in a New Keynesian model with search and matching frictions that closely follows Du (2024), which, in turn, is in spirit of the seminal work of Ravn and Sterk (2017, 2021). Aside from the household block of the model, the framework is standard and follows from the HANK and SAM literature. The model features nominal price rigidities à la Rotemberg, a monetary authority that sets the nominal interest rate following a standard Taylor rule that responds to inflation, and a fiscal authority that taxes labor income and borrows debt from households to fund unemployment insurance and interest on past debt. As in Gornemann, Kuester, and Nakajima (2021), Bardoczy (2024), and Graves (2024), households randomly search for jobs and match with a labor agency that sells labor to intermediate good producers. Complete details of the model are provided in appendix A.

The general equilibrium structure generates fiscal multipliers through an intertemporal Keynesian cross mechanism, which becomes particularly pronounced when monetary policy is passive. Moreover, the search and matching framework allows the employment rate to respond to policy interventions, allowing us to capture both demand and supply effects of fiscal policies.

Our approach in this section relies on linearizing the macro dynamics of the model and employs the Sequence Space Jacobian methods developed by Auclert, Bardóczy,

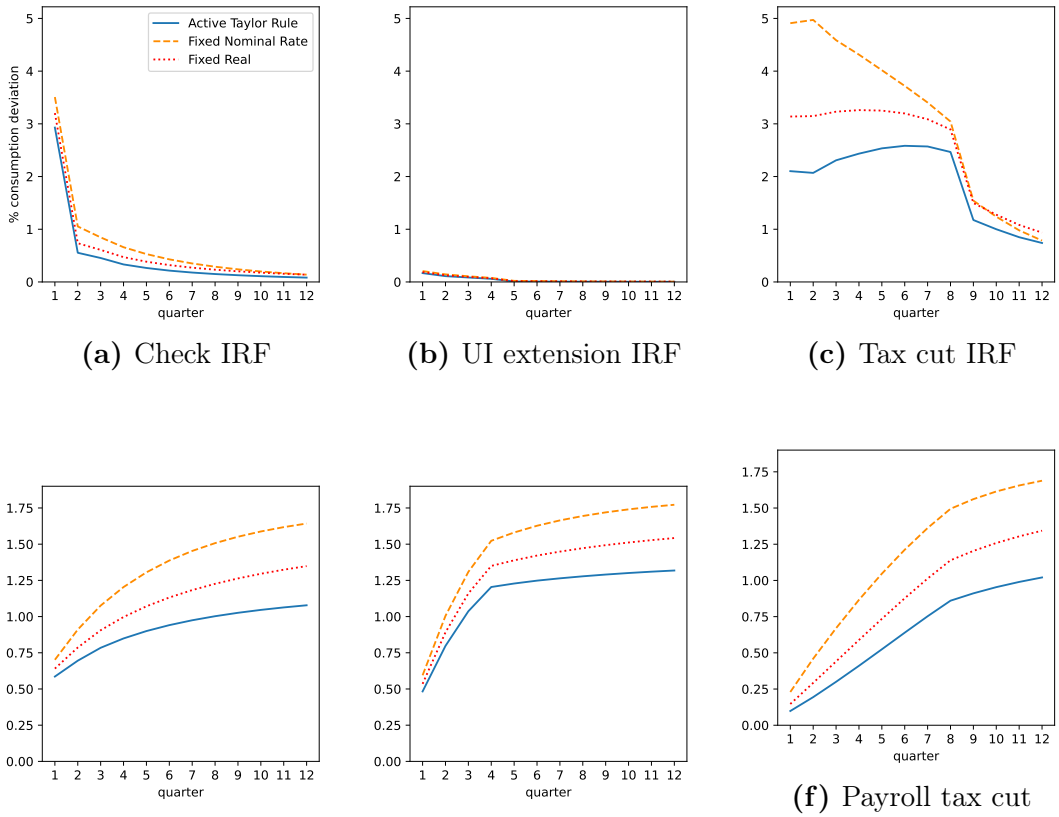
Rognlie, and Straub (2021). This linearization imposes certain constraints on our analysis. Notably, we cannot evaluate the effects of different policies starting from a deep recessionary state, as we do in our main results.[24] This limitation prevents us from conducting welfare comparisons between recessionary periods and the steady state. Additionally, the Keynesian cross mechanism embedded in the model exhibits uniform behavior regardless of the degree of economic slack——a feature that stands in contrast to the state-dependent multipliers we apply in our partial equilibrium analysis.[25]

The consumption response in this general equilibrium model to each of the three policies is shown in the top row of Figure 5. For each of the three fiscal policies, we have shown the consumption response under three different monetary policy rules: 1) an active Taylor rule with a coefficient of 1.5 on inflation; 2) a fixed nominal rate (simulating an effective lower bound); and 3) a fixed real rate (closest in spirit to our partial equilibrium analysis).

[24]One approach to overcome this limitation, which could be used in future work, is described in Boppart, Krusell, and Mitman (2018).

[25]We note two additional technical limitations of our general equilibrium implementation. First, stimulus payments in the model are specified as proportional to permanent income, rather than as means-tested fixed dollar amounts as implemented in practice and in our partial equilibrium framework. Second, splurge behavior only occurs out of equilibrium.

Figure 5 General equilibrium policy comparison in HANK-SAM model



(a) Check IRF **(b)** UI extension IRF **(c)** Tax cut IRF

(d) Check multiplier **(e)** UI extension multiplier **(f)** Payroll tax cut multiplier

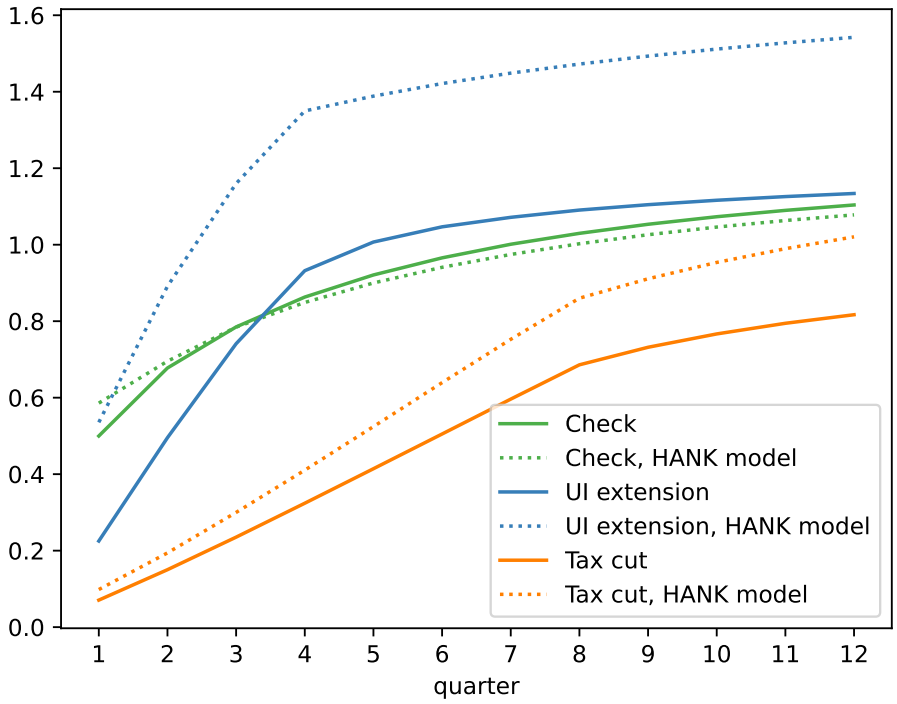
Note: This figure presents general equilibrium results from the HANK-SAM model (Section 5). Subfigures (a)-(c) show consumption impulse responses under three monetary policy rules: active Taylor rule, fixed nominal rate (ZLB), and fixed real rate. Subfigures (d)-(f) show corresponding cumulative multipliers over time. The general equilibrium framework generates multipliers through an intertemporal Keynesian cross mechanism, particularly pronounced when monetary policy is passive. Results validate the partial equilibrium findings that UI extensions achieve the highest multipliers across all policy environments.

Overall, the IRFs from this model are similar to those from the partial equilibrium analysis, especially under the fixed real-rate rule. Note that the magnitude of the consumption response to the UI extension is lower than in our main analysis—a consequence of lower long-term unemployment in this HANK exercise of deviating from the steady state.[26] Furthermore, although we are unable to repeat our welfare analysis under a recession in this model, the distributional effects of the policies are similar. Most importantly, the mechanism leading to far greater welfare benefits for the UI extension,

[26]By contrast, our main analysis considers deviations from a recessionary scenario. Note that the dynamics of the UI extension IRF are also somewhat faster acting. This is because, under the recession that we study in the partial equilibrium analysis, the large mass of newly-unemployed households do not start receiving extended UI for six months.

namely that the newly unemployed have high marginal utility, are robust to the supply-side effects of a general equilibrium HANK and SAM model.

Figure 6 Multiplier comparison: partial equilibrium vs. HANK model



Note: This figure compares consumption multipliers across model frameworks (Section 5). The comparison shows multipliers over different time horizons under a fixed real rate rule between the baseline partial equilibrium model (with state-dependent multipliers during recessions) and the general equilibrium HANK-SAM model (with uniform Keynesian cross mechanism). Both models show the tax cut policy substantially underperforming relative to stimulus checks and UI extensions, validating the robustness of policy rankings across modeling approaches. The HANK model produces larger overall multipliers due to general equilibrium effects.

The bottom row of Figure 5 shows the corresponding cumulative multipliers for each policy and monetary policy rule. Figure 6 compares these consumption multipliers over different horizons under a fixed real rate rule to those in our baseline partial equilibrium model. The multipliers are bigger in the HANK and SAM model. Nevertheless, in both models, the relative ranking of the consumption multipliers over time horizons are similar, with the effect of the tax cut substantially smaller than the stimulus check or UI extension policies, despite the inclusion of supply-side effects in this HANK model. However, in contrast to the partial equilibrium model, towards the end of the period shown the tax cut consumption multiplier is near that of the stimulus check. This is because the aggregate demand effects in our partial equilibrium model do not continue beyond the recession, dampening the benefits of the tax cut policy—in which much of

the extra spending occurs after the recession is over—relative to the stimulus check and extended UI policies.

As discussed in the literature review, Broer, Druedahl, Harmenberg, and Öberg (2025) also compute fiscal multipliers for commonly implemented stimulus policies in a HANK and SAM framework. A key distinction is that our model exhibits more spending out of income shocks in the quarters following the quarter of the shock. As shown in Auclert, Rognlie, and Straub (2018), accounting for the path of iMPCs beyond the first quarter significantly amplifies cumulative fiscal multipliers. By capturing this persistence—consistent with microeconomic evidence—our model produces larger multipliers for the untargeted stimulus check than those in Broer, Druedahl, Harmenberg, and Öberg (2025).

6 Conclusion

This paper studies the macroeconomic effects of both conventional (target-rate) and unconventional (forward-guidance, FG) monetary policy innovations in Mexico using high-frequency surprises around policy announcements. A central message is that raw interest rate surprises need not be clean measures of exogenous MP shocks: they can embed (i) CBI revealed about fundamentals and (ii) systematic RN components that reflect an imperfectly understood reaction function. When these informational components are not separated, contractionary target-rate shocks exhibit the familiar output, price, and exchange-rate puzzles and are followed by a counterintuitive rise in inflation expectations, while contractionary FG shocks appear to generate disproportionately large and fast real effects.

To address this issue, we combine projections on pre-announcement macro-financial information with high-frequency co-movements in equity prices to disentangle pure monetary policy innovations from RN and CBI shocks. Once CBI and RN components are isolated, the estimated effects of contractionary target-rate shocks become broadly consistent with standard transmission mechanisms, featuring delayed contractions in output and prices, an appreciation of the domestic currency, and well-anchored inflation expectations. For FG, isolating RN shocks substantially attenuates the apparent "forward guidance puzzle": the implied macroeconomic effects become more conservative and broadly similar to those of pure target-rate shocks.

A key contribution is that RN components play *asymmetric* contamination roles across policy instruments. In our estimates, RN shocks embedded in target-rate surprises tend to attenuate the contractionary effects of conventional MP shocks, while RN shocks embedded in FG surprises instead amplify and accelerate the effects of FG innovations. Beyond distinguishing conventional from unconventional tools, the results also speak to the exchange-rate puzzle in EMEs by highlighting how omitted informational components can distort estimated exchange-rate responses to policy tightenings.

Overall, the findings underscore that credible measurement of MP shocks in EMEs requires explicitly accounting for informational and systematic policy-response components in high-frequency surprises. Doing so yields impulse responses that are more in-

terpretable, more comparable across policy tools, and more consistent with conventional monetary transmission.

Appendices

A Details of the HANK and SAM Model

A.1 Households

The household block follows closely to the main text with a few exceptions. First, the splurge only occurs out of equilibrium—that is, the steady state of the model is calculated without the splurge behavior. Second, the level of permanent income of all newborns is equal to one. Furthermore, all households face the same employment to unemployment and unemployment to employment probabilities. The probabilities are calibrated to the transition probabilities of high school graduates from the main text. Lastly, following the notation of Auclert, Rognlie, and Straub (2020), r_t^a will denote the economy wide ex-ante real interest rate.

A.2 Goods Market

A continuum of monopolistically competitive intermediate goods producers, indexed by $j \in [0, 1]$, produces intermediate goods Y_{jt}, which are sold to a final goods producer at price P_{jt}. Each period, these producers fully consume their profits.

A.3 Final Goods Producer

A perfectly competitive final goods producer purchases intermediate goods Y_{jt} from intermediate good producer j at price P_{jt} and produces the final good Y_t using a CES production function:

$$Y_t = \left(\int_0^1 Y_{jt}^{\frac{\epsilon_p - 1}{\epsilon_p}} \, dj \right)^{\frac{\epsilon_p}{\epsilon_p - 1}},$$

where ϵ_p is the elasticity of substitution.

Given P_{jt}, the price of intermediate good j, the final goods producer maximizes profits by solving:

$$\max_{Y_{jt}} P_t \left(\int_0^1 Y_{jt}^{\frac{\epsilon_p - 1}{\epsilon_p}} \, dj \right)^{\frac{\epsilon_p}{\epsilon_p - 1}} - \int_0^1 P_{jt} Y_{jt}, \; dj.$$

The first order condition leads to demand for good j given by

$$Y_{jt} = \left(\frac{P_{jt}}{P_t} \right)^{-\epsilon_p} Y_t,$$

and the price index

$$P_t = \left(\int_0^1 P_{jt}^{1-\epsilon_p} \, dj \right)^{\frac{1}{1-\epsilon_p}}.$$

A.4 Intermediate Goods Producers

Intermediate goods producers produce according to a production function linear in labor L_t:

$$Y_{jt} = ZL_{jt},$$

where Z is total factor productivity.

Each intermediate goods producer hires labor L_t from a labor agency at cost h_t. Given this labor cost, each producer sets P_{jt} to maximize profits while facing price stickiness à la Rotemberg (1982). In HANK models with sticky prices, profits tend to be countercyclical, and when households have high MPCs, this can generate countercyclical consumption responses to dividends. To simplify, we assume that intermediate goods producers fully consume their profits rather than distributing them to households, thereby abstracting from consumption responses to firm profits. Each producer maximizes profits by solving:

$$J_t\left(P_{jt}\right) = \max_{\{P_{jt}\}} \left\{ \frac{P_{jt}Y_{jt}}{P_t} - h_t L_{jt} - \frac{\varphi}{2}\left(\frac{P_{jt} - P_{jt-1}}{P_{jt-1}}\right)^2 Y_t + J_{t+1}\left(P_{jt+1}\right) \right\},$$

where φ determines the cost of adjusting the price and, hence, the degree of price stickiness.

The problem can be rewritten as the standard New Keynesian maximization problem:

$$\max_{\{P_{jt}\}} \mathrm{E}_t \left[\sum_{s=0}^{\infty} M_{t,t+s} \left(\left(\frac{P_{jt+s}}{P_{t+s}} - MC_{t+s} \right) Y_{jt+s} - \frac{\varphi}{2}\left(\frac{P_{jt+s}}{P_{jt+s-1}} - 1 \right)^2 Y_{t+s} \right) \right],$$

where $MC_t = \frac{h_t}{Z}$.

Given that all firms face the same adjustment costs, there exists a symmetric equilibrium where all firms choose the same price with $P_{jt} = P_t$ and $Y_{jt} = Y_t$.

The resulting Phillips Curve is

$$\epsilon_p MC_t = \epsilon_p - 1 + \varphi(\Pi_t - 1)\Pi_t - M_{t,t+1}\varphi(\Pi_{t+1} - 1)\Pi_{t+1}\frac{Y_{t+1}}{Y_t}$$

where $\Pi_t = \frac{P_t}{P_{t+1}}$.

A.5 Labor market

A risk-neutral labor agency supplies labor N_t to intermediate goods producers at cost h_t by hiring households at wage w_t. To hire workers, the agency posts vacancies v_t, which are filled with probability ϕ_t. Household job search is random. Following Bardoczy (2024), we assume the labor agency cannot observe individual household productivity. Instead, it only observes the average productivity of all employed workers, which is normalized to one.

Labor agency. The labor agency determines how many vacancies to post and how much labor to sell by solving the following problem:

$$J_t(N_{t-1}) = \max_{N_t, v_t} \left\{ (h_t - w_t)N_t - \kappa v_t + \mathrm{E_t}\left[\frac{J_{t+1}(N_t)}{1 + r_t^a}\right] \right\},$$

subject to

$$N_t = (1-\omega)N_{t-1} + \phi_t v_t.$$

The parameters κ and ω are, respectively, the cost of posting a vacancy, and the job separation rate.

The resulting job creation curve is:

$$\frac{\kappa}{\phi_t} = (h_t - w_t) + (1-\omega)\mathrm{E_t}\left[\frac{\kappa}{(1+r_t^a)\phi_{t+1}}\right].$$

Matching. The matching process between households and the labor agency follows a Cobb-Douglas matching function:

$$m_t = \chi e_t^{\alpha} v_t^{1-\alpha},$$

where m_t is the mass of matches, e_t is the mass of job searchers, α is the matching function elasticity, and χ is a matching efficiency parameter.

The vacancy filling probability ϕ_t and the job finding probability η_t evolve as follows:

$$\eta_t = \chi \Theta_{it}^{1-\alpha}$$

$$\phi_t = \chi \Theta_t^{-\alpha}$$

where $\Theta_t = \frac{v_t}{e_t}$ is labor market tightness.

Wage Determination. Following Gornemann, Kuester, and Nakajima (2021) and Blanchard and Galí (2010), we assume the real wage evolves according to the following rule:

$$\log\left(\frac{w_t}{w_{ss}}\right) = \phi_w \log\left(\frac{w_{t-1}}{w_{ss}}\right) + (1-\phi_w)\log\left(\frac{N_t}{N_{ss}}\right),$$

where ϕ_w dictates the extent of real wage rigidity.

A.6 Fiscal Policy

The government issues long term bonds B_t at price q_t^b in period t that pays δ^s in period $t+s+1$ for $s \in \{0, 1, 2, \ldots\}$.

The bond price satisfies the no arbitrage condition:

$$q_t^b = \frac{1 + \delta \mathrm{E}_t[q_{t+1}^b]}{1 + r_t^a}.$$

The government funds its expenditures through debt and taxes, subject to the following budget constraint:

$$(1 + \delta q_t^b)B_{t-1} + G_t + S_t = \tau_t w_t N_t + q_t^b B_t,$$

where S_t are payments for unemployment insurance and other transfers.

For all stimulus policies, except tax cuts, we follow Auclert, Rognlie, and Straub (2020) and allow the tax rate to adjust in order to stabilize the debt-to-GDP ratio:

$$\tau_t - \tau_{ss} = \phi_B q_{ss}^b \frac{B_{t-1} - B_{ss}}{Y_{ss}}$$

where ϕ_B governs the speed of adjustment.

For the tax cuts, we assume government expenditures adjust following:

$$G_t - G_{ss} = \phi_G q_{ss}^b \frac{B_{t-1} - B_{ss}}{Y_{ss}}$$

where ϕ_G governs the speed of adjustment of government spending in response to debt.

A.7 Monetary Policy

The central bank follows a standard Taylor rule that responds solely to inflation:

$$i_t = r^* + \phi_\pi \pi_t,$$

where ϕ_π is the coefficient on inflation. Inflation is given by $\pi_t = P_t/P_{t-1} - 1$, and r^* is the steady state interest rate.

A.8 Equilibrium

An equilibrium in this economy is a sequence of:

- Policy Functions $(c_{it}(m))_{t=0}^{\infty}$ normalized by permanent income.
- Prices $\left(r_{t+1}^a, i_t, q_t^b, w_t, h_t, \pi_t, \tau_t\right)_{t=0}^{\infty}$.
- Aggregates $(C_t, Y_t, N_t, \Theta_t, B_t, A_t)_{t=0}^{\infty}$.

Such that:

- $(c_{it}(m))_{t=0}^{\infty}$ solves the household's maximization problem given $(w_t, \eta_t, r_t^a, \tau_t)_{t=0}^{\infty}$.
- The final goods producer and intermediate goods producers both maximize their respective objective functions.
- The nominal interest rate is determined by the central bank's Taylor rule.

Table 3 Calibration

Description	Parameter	Value	Source/Target
Elasticity of Substitution	ϵ_p	6	Standard
Price Adjustment Costs	φ	96.9	Ravn and Sterk (2017, 2021)
Vacancy Cost	κ	0.056	$\frac{\kappa}{w\phi} = 0.071$
Job Separation Rate	ω	0.092	Match $\pi(eu)$ for Highschool group
Matching Elasticity	α	0.65	Ravn and Sterk (2017, 2021)
Job Finding Probability	η_{ss}	0.67	$\pi(ue)$ in section **??**
Vacancy Filling Rate	ϕ_{ss}	0.71	den Haan et al. (2000)
Real Wage Rigidity parameter	ϕ_w	0.837	Gornemann et al. (2021)
Government Spending	G	0.38	Gov. budget constraint
Decay rate of Gov. Coupons	δ	0.95	5 Year Maturity of Debt
Response of Tax Rate to Debt	ϕ_B	0.015	Auclert et al. (2020)
Taylor Rule Inflation Coefficient	ϕ_π	1.5	Standard

- The tax rate is set according to the fiscal rule, ensuring that the government budget constraint is satisfied.
- The value of assets is equal to the value of government bonds:

$$A_t = q_t^b B_t.$$

- The goods market clears[27]:

$$C_t = w_t N_t + G_t,$$

  where $C_t \equiv \int_0^1 \mathbf{c}_{it}\, di$.
- The labor demand of intermediate goods producers equals labor supply of labor agency:

$$L_t = N_t.$$

[27]Note if profits were not held by firms then the goods market condition would be $C_t + G_t = Y_t - \kappa v_t - \frac{\varphi}{2}\left(\frac{P_t}{P_{t-1}} - 1\right)^2 Y_t$. In particular, since firm profits are $D_t = Y_t - w_t N_t - \kappa v_t - \frac{\varphi}{2}\left(\frac{P_t}{P_{t-1}} - 1\right)^2 Y_t$, then the goods market condition would become $C_t + G_t = w_t N_t + D_t = Y_t - \kappa v_t - \frac{\varphi}{2}\left(\frac{P_t}{P_{t-1}} - 1\right)^2 Y_t$.

A.9 Calibration of Non-Household Blocks

The elasticity of substitution is set to 6, and the price adjustment cost parameter is set to 96.9 as in Ravn and Sterk (2017, 2021). The vacancy cost is set to 7 percent of the real wage as in Christiano, Eichenbaum, and Trabandt (2016).[28] The matching elasticity is 0.65 following Ravn and Sterk (2017, 2021). The job separation rate is set to 0.092. As in section **??**, we set the job finding probability in the steady state for the unemployed η_{ss} to 0.67. Along with the job separation rate, this gives a probability of transitioning from employment to unemployment within a quarter of 3.1 percent which is the value we use for the Highschool group in section **??**. The quarterly vacancy filling rate is 0.71 as in den Haan, Ramey, and Watson (2000) (and together with our other choices, this pins down the matching efficiency χ). The degree of wage rigidity ϕ_w is set to 0.837 following Gornemann, Kuester, and Nakajima (2021). The tax rate is set to 0.3 and government spending is set to clear the government budget constraint. The parameters that dictate the speed of fiscal adjustment, ϕ_B and ϕ_G, are set to 0.015, the lower bound of the estimates in Auclert, Rognlie, and Straub (2020).[29] Furthermore, the decay rate of government coupons is set to $\delta = 0.95$ to match a maturity of 5 years.[30] Finally, the Taylor rule coefficient on inflation is set to the standard value of $\phi_\pi = 1.5$.

[28] The range of plausible values lie between 4*percent* and 14*percent* Silva and Toledo (2009)

[29] The speed of adjustment parameter is set to the lower bound to ensure that the policies evaluated in the HANK and SAM model are almost entirely deficit financed.

[30] The duration of bonds in the model is $\frac{(1+r)^4}{(1+r)^4-\delta}$

B Results in a Model Without the Splurge

Full appendix PDF available at: `GitHub: Subfiles/Appendix-NoSplurge.pdf`

Click any section title below to view it online in HTML:

B.1 Introduction

B.2 Matching the iMPCs without the splurge

B.3 Estimating discount factor distributions without the splurge

B.4 Multipliers in the absence of the splurge

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